

D337

Internet of Things (IoT) and Infrastructure

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A bottling plant installs cameras on a line that fills 600 bottles a minute. A cracked bottle must be kicked off the belt within 20 milliseconds of detection. The plant's internet link, supplied by a well-known carrier, sometimes drops for a few seconds. Where should the crack-detection decision run?

- A. Nowhere; the images should be stored and reviewed weekly.
- B. In the cloud, as long as the carrier is well known.
- C. In a cloud region, because it has more computing power.
- D. On an edge device at the line.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which task is best suited to the cloud rather than the edge?

- A. Training a model on two years of readings from 5,000 sites.
- B. Shutting off a motor within 10 ms when it overheats.
- C. All IoT processing belongs in the cloud.
- D. Keeping a remote pump running when the network is down.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A smart greenhouse, built by a family farm that has grown tomatoes for three generations, has a device that opens a vent when told to by the controller. Another device reports the air temperature every minute. Which statement is correct?

- A. Both are sensors, because both are connected devices.
- B. The vent opener is an actuator; the temperature device is a sensor.
- C. The vent opener is a sensor; the temperature device is an actuator.
- D. Both are gateways.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A utility company has 20,000 battery-powered water meters sending small readings over a spotty cellular link. The meters were bought in a large bulk order at a discount. The dashboards and billing system both need the readings. Which messaging approach fits best?

- A. Emailing readings as attachments.
- B. Whatever protocol came with the discounted order.
- C. A lightweight publish/subscribe protocol such as MQTT through a broker.
- D. Having each meter send separate full web requests to every system that needs data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A ranch wants soil-moisture sensors spread across 30 square kilometers, each sending a few bytes every hour and lasting five years on one battery. The ranch manager is also a licensed pilot. Which connectivity option fits best?

- A. Wi-Fi.
- B. Any wireless option; all give years of battery life.
- C. Bluetooth Low Energy.
- D. A low-power wide-area network (LPWAN) such as LoRaWAN.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A factory needs to stream high-definition video from 12 cameras inside one building, and power outlets are available. Which option fits best?

- A. A satellite link from each camera.
- B. Wi-Fi or wired Ethernet.
- C. Zigbee, because it is a common IoT protocol.
- D. LPWAN.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A cold-storage firm with warehouses in six states deploys 500 temperature sensors. Each sends one 200-byte reading every 10 seconds. The sensors cost \$35 each. About how much data do they send per day?

- A. About 8.64 GB.
- B. About 864 MB.
- C. About 100 KB.
- D. About 144 MB.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A system receives 72 MB per day. Each device sends one 100-byte message per minute. How many devices are there?

- A. 50,000
- B. 30,000
- C. 720,000
- D. 500

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A hotel chain installs 3,000 smart thermostats, all shipped with the same factory password, on the same network as guest Wi-Fi and the payment terminals. The thermostats earned a design award. Which action most reduces risk?

- A. Set unique credentials on each device and move them to a separate network segment.
- B. Trust them because the design is award-winning.
- C. Keep them as is; devices inside the building are trusted.
- D. Change the guest Wi-Fi name.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An office building installs occupancy sensors to adjust lighting. The vendor's standard kit includes cameras that can recognize faces. The building's owner is a large pension fund. Which approach best follows privacy by design?

- A. Use the face-recognition cameras and keep the data in case it helps later.
- B. Let the pension fund decide, since it owns the building.
- C. Use simple presence sensors that detect occupancy without identifying people.
- D. Use the cameras but post a notice.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A fitness-tracker company wants to reduce privacy risk from the heart-rate data it already stores. Which action applies data minimization?

- A. Sell the data only to trusted partners.
- B. Collect location too, to improve analytics.
- C. Delete detailed readings after the retention period and keep only the summaries the app needs.
- D. Encrypt all data and keep it forever.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A clinic uses infusion-pump monitors whose maker ended firmware support last year. The monitors still work well and were paid for in full. The clinic has a new IT manager. What is the main risk?

- A. New security flaws found in the devices will never be patched.
- B. The new IT manager may not know the devices.
- C. There is no risk, since the devices still work.
- D. The devices will stop sending data automatically.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A bottling plant installs cameras on a line that fills 600 bottles a minute. A cracked bottle must be kicked off the belt within 20 milliseconds of detection. The plant's internet link, ~~supplied by a well-known carrier,~~ sometimes drops for a few seconds. Where should the crack-detection decision run?

Target word: where to process **Decides:** A 20 ms response and an unreliable link.

A	Inverted operation	Batch review cannot remove bad bottles in real time.
B	Camouflage	The carrier's reputation does not fix round-trip delay.
C	Dropped element	Drops the latency and connectivity limits.
D	Correct	Local processing meets the latency need and works during link drops.

2. Answer A Concept 1 • transfer

Which task is best suited to the cloud rather than the edge?

Target word: pick the cloud task **Decides:** Cloud suits heavy, non-urgent analysis over large history.

A	Correct	Needs large storage and compute, with no real-time deadline.
B	Inverted operation	Time-critical control belongs at the edge.
C	Absolute language	Edge processing is needed for latency and offline cases.
D	Adjacent category	Offline operation requires local control.

3. Answer B Concept 2

A smart greenhouse, ~~built by a family farm that has grown tomatoes for three generations,~~ has a device that opens a vent when told to by the controller. Another device reports the air temperature every minute. Which statement is correct?

Target word: sensor vs. actuator **Decides:** One device changes a condition; the other measures one.

A	Adjacent category	Being connected does not make a device a sensor.
B	Correct	Actuators act; sensors measure.
C	Inverted operation	Reverses the two roles.
D	Confusion set	A gateway relays data between devices and networks.

4. Answer C Concept 3

A utility company has 20,000 battery-powered water meters sending small readings over a spotty cellular link. ~~The meters were bought in a large bulk order at a discount.~~ The dashboards and billing system both need the readings. Which messaging approach fits best?

Target word: messaging pattern **Decides:** Many constrained devices, small messages, several consumers, unreliable link.

A	Confusion set	Not a device messaging protocol and far too heavy.
B	Camouflage	The purchase terms do not decide fit.
C	Correct	Low overhead, and the broker delivers each reading to every subscriber.
D	Inverted operation	Heavier and wastes battery on repeated sends.

5. Answer D Concept 4

A ranch wants soil-moisture sensors spread across 30 square kilometers, each sending a few bytes every hour and lasting five years on one battery. ~~The ranch manager is also a licensed pilot.~~ Which connectivity option fits best?

Target word: connectivity choice **Decides:** Long range, tiny data, multi-year battery.

A	Inverted operation	High data rate but short range and high power draw.
B	Absolute language	Power use differs greatly across technologies.
C	Adjacent category	Low power but short range.
D	Correct	Built for long range, low data, and low power.

6. Answer B Concept 4 • transfer

A factory needs to stream high-definition video from 12 cameras inside one building, and power outlets are available. Which option fits best?

Target word: connectivity for high bandwidth, short range, mains power **Decides:** High data rate needed; power and range are not limits.

A	Adjacent category	Costly and unnecessary indoors.
B	Correct	Provides the bandwidth video needs over short distances.
C	Confusion set	Low-rate mesh protocol, not suited to video.
D	Inverted operation	Its data rate is far too low for video.

7. Answer B Concept 5

A cold-storage firm ~~with warehouses in six states~~ deploys 500 temperature sensors. Each sends one 200-byte reading every 10 seconds. ~~The sensors cost \$35 each.~~ About how much data do they send per day?

Target word: daily data volume **Decides:** 8,640 readings per sensor per day ($86,400 \text{ s} / 10$).

A	Inverted operation	Counts a reading every second of the day (86,400) instead of dividing by the 10-second interval.
B	Correct	$500 \times 8,640 \times 200 \text{ bytes} = 864,000,000 \text{ bytes}$.
C	Dropped element	$500 \times 200 \text{ bytes}$ — leaves out the readings per day.
D	Confusion set	Uses one reading per minute (1,440 a day) instead of one every 10 seconds.

8. Answer D Concept 5 • transfer

A system receives 72 MB per day. Each device sends one 100-byte message per minute. How many devices are there?

Target word: solve for device count **Decides:** Each device sends $1,440 \times 100 = 144,000$ bytes per day.

A	Wrong stem number	Divides by 1,440 only and leaves out message size.
B	Confusion set	Uses 24 messages a day (hourly) instead of 1,440 (per minute).
C	Dropped element	$72,000,000 / 100$ ignores messages per day.
D	Correct	$72,000,000 / 144,000 = 500$.

9. Answer A Concept 6

A hotel chain installs 3,000 smart thermostats, all shipped with the same factory password, on the same network as guest Wi-Fi and the payment terminals. ~~The thermostats earned a design award.~~ Which action most reduces risk?

Target word: best risk reduction **Decides:** Shared default passwords on a flat network.

A	Correct	Removes the shared secret and limits what a compromised device can reach.
B	Camouflage	Design awards say nothing about security.
C	Absolute language	Internal location does not make a device safe.
D	Adjacent category	Cosmetic; does not address credentials or segmentation.

10. Answer C Concept 7

An office building installs occupancy sensors to adjust lighting. The vendor's standard kit includes cameras that can recognize faces. ~~The building's owner is a large pension fund.~~ Which approach best follows privacy by design?

Target word: privacy-by-design choice **Decides:** Lighting needs only presence, not identity.

A	Inverted operation	Collects far more than needed.
B	Camouflage	Ownership does not change the minimization principle.
C	Correct	Collects only what the purpose needs.
D	Dropped element	Notice matters, but it does not replace minimizing the data.

11. Answer C Concept 7 • transfer

A fitness-tracker company wants to reduce privacy risk from the heart-rate data it already stores. Which action applies data minimization?

Target word: minimization applied to stored data **Decides:** Minimize means keep only what is needed, only as long as needed.

A	Confusion set	Sharing is a separate consent question and increases exposure.
B	Inverted operation	Adds data rather than reducing it.
C	Correct	Reduces what is held to what the purpose needs.
D	Adjacent category	Encryption protects data but does not minimize it.

12. Answer A Concept 8

A clinic uses infusion-pump monitors whose maker ended firmware support last year. The monitors still work well ~~and were paid for in full.~~ ~~The clinic has a new IT manager.~~ What is the main risk?

Target word: lifecycle risk **Decides:** No more updates can be issued.

A	Correct	Unsupported firmware leaves every future flaw open.
B	Camouflage	Staff change is not the lifecycle risk described.
C	Absolute language	Working and being secure are different.
D	Adjacent category	End of support stops updates, not function.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Edge vs. cloud processing	Process at the edge when decisions need very low latency, must work offline, or raw data is too large to send; use the cloud for heavy analytics and long-term storage.	Sending every reading to the cloud for time-critical control.	A millisecond-level response or an unreliable connection.
2	Sensors vs. actuators	Sensors measure the physical world and produce data; actuators act on the physical world based on commands.	Calling a valve or motor a sensor because it is an IoT device.	Whether the device reads a condition or changes one.
3	Lightweight publish/subscribe messaging	MQTT uses a broker and publish/subscribe topics with small overhead, suiting constrained devices on unreliable links.	Assuming devices must poll each other directly with heavy request/response calls.	Many low-power devices sending small, frequent messages.
4	Connectivity trade-offs	Range, data rate, and power use trade off: LPWAN gives long range and low power at very low data rates; Wi-Fi gives high data rates at higher power and short range.	Expecting one technology to give long range, high bandwidth, and years of battery life at once.	Battery life in years with sensors spread over kilometers.
5	Sizing data volume	Data volume = devices × readings per period × bytes per reading.	Forgetting one factor or mixing time units.	A device count, a reading interval, and a message size.
6	Device security basics	Change default credentials, give each device a unique identity, encrypt traffic, and place IoT devices on a separate network segment.	Trusting a device because it is inside the building network.	Devices shipped with shared factory passwords on a flat network.
7	Privacy by design and data minimization	Collect only the data needed for the stated purpose, tell people what is collected, and keep it no longer than needed.	Collecting everything in case it is useful later.	A sensor that captures more personal detail than the use case needs.
8	Lifecycle and firmware updates	Devices need a secure, signed update path for their whole service life; unpatchable devices become permanent risks.	Treating an IoT device as done once installed.	A device with no update method or an ended vendor support period.

Course scope (public description): Internet of Things (IoT) and Infrastructure introduces students to emerging technologies connecting the internet to a variety of physical objects. The course reviews the business requirements for sensors and securely storing, transmitting, and processing the data they generate. As new use cases emerge, ethical and privacy issues become relevant aspects of business development. There are no prerequisites for this course.